

Question 1:

Micro-Processor:

is an IC which can perform arithmetic and logical operations. It has only a CPU and it consists of ALU, control unit and registers.

Micro-Controller:

is an IC which consists of CPU and other components like ROM, Ram and other peripherals. Mainly used in embedded systems. It always known as small computer.

Embedded System:

Is a special purpose computer system design to do one or few number of tasks. Has some constrained in time, code size and power consumption.

Mechatronic System:

System in which mechanical hardware is integrated with information – drive system (Micro-Controllers)

n-bit processor:

The size of the data bus is n bits. That's mean it can handle n-bit of data at a time. The larger data has broken into pieces.

Question2:

<u>Micro-processor</u>	<u>Micro-controller</u>
General purpose	Specific purpose – include micro-processor with some other peripherals
Have no RAM, ROM or I/O peripherals	Have RAM, ROM and other peripherals connected with Micro-processor using buses
Disadvantages: 1. Harder to use because it requires connecting external peripherals to it to can be used in embedded systems. 2. More expensive than micro-controller	Disadvantages: 1. We can not add any external features such as add timer or I/O ports
Advantages: 1. It can enable the designer to design its own micro-controller with needed peripherals for the task.	Advantage: 1. Low cost and easy to use.

Question3:

Von-Neuman:

1. Single data bus for both instructions and data.
2. Can not use the pipeline technique.

Harvard architecture:

1. Separate buses for data and instructions.
2. Can use the pipeline technique.

Question4:

Types of ROM:

1. PROM
2. EPROM
3. EEPROM
4. Flash Memory EPROM

Question5:

Types of RAM:

1. Static Ram
2. Dynamic Ram

Question6:

It's named ROM sine the software can not write on it. But now with flash ROM and EEPROM we can write on it and read from it.

Question7:

Type	Volatile	Writable	Erase size	Max Erase Cycle	Cost / bite	Speed
SRAM	Yes	Yes	Byte	Infinity	Expensive	Fast
DRAM	Yes	Yes	Byte	Infinity	Moderate	Moderate
Masked ROM	No	No	N/A	N/A	Cheap	Fast
PROM	No	Once	N/A	N/A	Moderate	Fast
EPROM	No	Yes	All chip	Limited	Moderate	Fast
EEPROM	No	Yes	Byte	Limited	Expensive	Fast read slow write
FLASH	No	Yes	Sector	Limited	Moderate	Fast read slow write
NVRAM	No	Yes	Byte	infinity	Expensive	Fast